CBSE Class 12 Physics — Electromagnetic Waves

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark Questions (MCQ/Assertion-Reason/VSA)

Q1. Which of the following electromagnetic waves has the highest frequency? (a) X-ray (b) Ultraviolet rays (c) Infrared rays (d) Gamma rays [PYQ 2022] Very High

Q2. Displacement current exists only when: (a) electric field is changing. (b) magnetic field is changing. (c) electric field is not changing. (d) magnetic field is not changing. [PYQ 2020] Very High

Q3. A welder wears special glasses to protect his eyes mostly from the harmful effect of: (a) very intense visible light (b) infrared radiation (c) ultraviolet rays (d) microwaves [PYQ 2020] High

Q4. Electromagnetic waves used as a diagnostic tool in medicine are: (a) X-rays (b) ultraviolet rays (c) infrared radiation (d) ultrasonic waves [PYQ 2020] Very High

Q5. Choose the electromagnetic wave relevant to aircraft navigation: (a) ultraviolet (b) infrared (c) microwave (d) visible light [PYQ 2022] Very High

Q6. Which of the following laws was modified by Maxwell by introducing the concept of displacement current? (a) Gauss's law (b) Ampere's circuital law (c) Biot-Savart's law (d) Faraday's law [PYQ 2022] Very High

Q7. Which of the following statements is NOT true about the properties of electromagnetic waves? (a) These waves do not require any material medium for their propagation. (b) Both electric and magnetic field vectors attain the maxima and minima at the same time. (c) The energy in an electromagnetic wave is divided equally between electric and magnetic fields. (d) Both electric and magnetic field vectors are parallel to each other. [PYQ 2022] Very High

Q8. Give the ratio of the velocity of two light waves of wavelengths $4000 \ \backslash\mathrm{AA}$ and $8000 \ \backslash\mathrm{AA}$ travelling in a vacuum. [PYQ 2021] Very High

Q9. The charging current for a capacitor is 0.25 A. What is the magnitude of the displacement current across its plates? [PYQ 2016] High

Q10. Mention one use of the part of the electromagnetic spectrum to which a wavelength of 21 cm (emitted by hydrogen in interstellar space) belongs. [PYQ 2021] Medium

Q11. Write the mathematical expression for the speed of light in a material medium of relative permittivity ϵ_r and relative magnetic permeability μ_r. [PYQ 2020] High

Q12. If the electric field $\vec{E}$ of an electromagnetic wave is oscillating along the y-axis and the wave is propagating along the +x-axis, what is the direction of the oscillating magnetic field $\vec{B}$? [PYQ 2020] Very High

Q13. For an electromagnetic wave propagating in a vacuum, what does the ratio of its angular wave number k to its angular frequency ω represent? [PYQ 2025-26 SQP] High

Q14. Name the electromagnetic waves which are typically referred to as "heat waves". [PYQ 2022] Very High

Q15. Which electromagnetic waves lie near the high-frequency end of the visible part of the electromagnetic spectrum? [PYQ 2016] Medium

Q16. What is the dimensional formula of the ratio of the amplitude of the electric field to the amplitude of the magnetic field (E_0/B_0)? [PYQ 2025-26 SQP] High

Q17. Assertion (A): In an electromagnetic wave, the electric and magnetic fields vary with the same frequency and are in exactly the same phase. **Reason (R):** The energy associated with the electric field is much greater than that associated with the magnetic field. [PYQ 2025-26 SQP] Very High

Q18. Assertion (A): The displacement current flows in the dielectric of a capacitor when the potential difference across its plates is decreasing with time. **Reason (R):** A changing electric field produces a magnetic field. [PYQ 2025-26 SQP] High

Q19. Assertion (A): Microwaves are considered highly suitable for radar systems used in aircraft navigation. **Reason (R):** Microwaves have shorter wavelengths compared to radio waves, allowing them to travel as directed beams with minimal diffraction. [Practice] Very High

Q20. Assertion (A): Electromagnetic waves exert radiation pressure on surfaces they strike. **Reason (R):** Electromagnetic waves carry both energy and linear momentum. [PYQ 2019] Very High

2-Mark Questions

Q21. Electromagnetic waves with wavelength λ_1 are suitable for radar systems. Wavelength λ_2 is used to kill germs in water purifiers. Wavelength λ_3 is used to improve visibility on runways during fog. Identify and name the part of the electromagnetic spectrum to which these radiations belong. Arrange these wavelengths in ascending order of their magnitude. [PYQ 2022] Very High

Q22. How is a displacement current produced between the plates of a parallel plate capacitor during charging? Write the expression for it. [PYQ 2020] Very High

Q23. Depict the field diagram of an electromagnetic wave propagating along the positive x-axis with its electric field oscillating along the y-axis. [PYQ 2020] High

Q24. Why is the thin ozone layer on top of the stratosphere crucial for human survival? Identify the specific part of the electromagnetic spectrum it absorbs. [PYQ 2019] Very High

Q25. Identify the part of the electromagnetic spectrum used in (i) radar, and (ii) eye surgery (LASIK). Write their approximate frequency ranges. [PYQ 2019] High

Q26. State two basic properties of electromagnetic waves. How can we mathematically show that EM waves carry momentum? [PYQ 2016] Very High

Q27. Identify the electromagnetic waves whose wavelengths vary as: (a) $10^{-12}\ \mathrm{m} < \lambda < 10^{-8}\ \mathrm{m}$ and (b) $10^{-3}\ \mathrm{m} < \lambda < 10^{-1}\ \mathrm{m}$. Write one common use for each. [PYQ 2017] High

Q28. How are electromagnetic waves produced by oscillating charges? Briefly explain the energy conversion taking place. [PYQ 2016] High

Q29. Why are infrared waves referred to as heat waves? How are they produced? What role do they play in maintaining the earth's warmth? [PYQ 2015] Very High

Q30. How are microwaves produced? Why is it necessary in microwave ovens to select the frequency of microwaves to specifically match the resonant frequency of water molecules? [PYQ 2014] Very High

Q31. Prove mathematically that the average energy density of the oscillating electric field is exactly equal to that of the oscillating magnetic field in an electromagnetic wave. [PYQ 2019] High

Q32. An electromagnetic wave is travelling in a medium with velocity v along the x-axis. How are the magnitudes of the electric and magnetic fields related to the velocity of the EM wave? Give the mathematical expression. [PYQ 2016] Medium

3-Mark Questions

Q33. Write Maxwell's generalization of Ampere's Circuital law. Show that in the process of charging a capacitor, the displacement current produced between the plates is $I_d = \epsilon_0(d\phi_E/dt)$, where ϕ_E is the electric flux. [PYQ 2016] Very High

Q34. An electromagnetic wave of frequency $1\ \mathrm{GHz}$ travels through empty space along the z-direction. At a specific point in space, the electric field $\vec{E}$ attains a maximum value of $50\ \mathrm{V/m}$. If the electric wave is polarized along the x-axis, (a) identify the plane in which the magnetic field $\vec{B}$ will lie, and (b) express the electric and magnetic fields as a function of z and t. [PYQ 2025-26 SQP] Very High

Q35. A dish antenna with a circular aperture of radius $20\ \mathrm{cm}$ receives digital TV signals from a satellite. The average intensity of the EM waves carrying the signal is $5 \times 10^{-14}\ \mathrm{W/m^2}$. Determine: (a) the total electromagnetic energy delivered to the dish during a 30-minute telecast, and (b) the average energy density of the incoming EM wave. [PYQ 2025-26 SQP] High

Q36. A satellite at a height of $100\ \mathrm{km}$ from the Earth's surface detects a radio signal uniformly emitted in all upward directions by a radio station on the ground. If the average power of the signal emitted is $100\ \mathrm{kW}$, find the amplitudes of the electric field (E_0) and magnetic field (B_0) of the incoming signal at the satellite. [PYQ 2025-26 SQP] Very High

Q37. In a spaceship orbiting the Earth, two astronauts stationed $2\ \mathrm{m}$ apart speak to each other via sound waves. Their conversation is simultaneously transmitted to Earth via electromagnetic waves. If the Earth station is $1765\ \mathrm{km}$ away, calculate the respective times taken for the sound waves to travel between the astronauts and the EM waves to reach Earth. Compare them. (Take speed of sound = $340\ \mathrm{m/s}$). [PYQ 2025-26 SQP] Medium

Q38. A laser emits a sinusoidal EM wave of wavelength $10\mu\mathrm{m}$ along the +x-axis. The electric field $\vec{E}$ of the wave is parallel to the negative z-axis with a maximum amplitude of $1\ \mathrm{MV/m}$. Express the wave equations for both $\vec{E}$ and $\vec{B}$ for this wave, substituting all appropriate numerical values and unit vectors. [PYQ 2025-26 SQP] High

Q39. The cosmic microwave background radiation throughout the universe has an average energy density of roughly $4 \times 10^{-14}\ \mathrm{J/m^3}$. Calculate the rms value of the electric field (E_{rms}) associated with this radiation. Also, mathematically relate the rms electric field to the rms magnetic field. [PYQ 2025-26 SQP] Medium

Q40. Which segment of the electromagnetic spectrum has the highest frequency? How are these waves naturally produced? Write two of their most significant practical applications in medicine or industry. [PYQ 2016] Very High

Q41. Light of uniform intensity shines perpendicularly on a totally absorbing surface. (a) Derive an expression for the radiation pressure exerted on the surface. (b) If the surface area is decreased while keeping the light source identical, does the intensity change? Explain. [PYQ 2025-26 SQP] High

Q42. Explain the fundamental inconsistency in Ampere's circuital law when applied to a charging parallel plate capacitor, which led James Clerk Maxwell to introduce the concept of displacement current. [Practice] Very High

5-Mark Questions (Long Answer/Case-Study)

Q43. Case Study: Maxwell's Displacement Current When a parallel plate capacitor is being charged, an ammeter in the external circuit shows a current, implying charge is flowing. However, between the capacitor plates, there is a physical gap with no conducting medium. James Clerk Maxwell deduced that a changing electric field in this gap produces an effect identical to a physical current, which he termed 'displacement current'. (a) Define displacement current and write its mathematical formula. (b) How does the introduction of displacement current mathematically resolve the inconsistency in Ampere's circuital law? Write the modified Ampere-Maxwell law. (c) A parallel plate capacitor with circular plates of radius R is being charged by a steady current I. Find the magnetic field B at a radial distance r from the central axis between the plates for $r < R$. (d) Does the displacement current persist after the capacitor is fully charged? Explain. [Practice] Very High

Q44. Case Study: The Electromagnetic Spectrum The electromagnetic spectrum spans an enormous range of frequencies, from long radio waves to ultra-short gamma rays. Different frequency bands are produced by different physical mechanisms and interact with matter in distinct ways, dictating their practical applications. Identify the specific electromagnetic waves described by the following applications and state their approximate frequency or wavelength band: (a) Used in cellular networks and Wi-Fi communication. (b) Used by security scanners at airports to see through luggage. (c) Emitted by radioactive nuclei and used to sterilize medical equipment. (d) Detected by night-vision goggles and thermal imaging cameras. (e) Used by plants for photosynthesis and perceived by the human eye. [Practice] High

Q45. (a) Discuss the transverse nature of electromagnetic waves. How do the electric and magnetic fields orient themselves relative to each other and the direction of propagation? (b) A plane electromagnetic wave travels in a vacuum along the z-direction. What can you say about the directions of its electric and magnetic field vectors? (c) If the magnitude of the magnetic field in a plane EM wave is $B_z = 2 \times 10^{-7} \sin(1.5 \times 10^3 x + 4.5 \times 10^{11} t)\ \mathrm{T}$, find the (i) wavelength, (ii) frequency, and (iii) the expression for the corresponding electric field. [Practice] Very High

Q46. (a) What is radiation pressure? Write the expressions for the momentum transferred to a surface by an EM wave of energy U when the surface is (i) totally absorbing, and (ii) perfectly reflecting. (b) A $20\ \mathrm{W}$ laser beam shines perpendicularly on a completely absorbing black surface. Calculate the force exerted by the light on the surface. (c) Why do we not normally feel the radiation pressure of sunlight on our bodies? [Practice] High

Q47. (a) Explain the term 'Electromagnetic Wave'. Write the four fundamental Maxwell's equations in integral form. (b) Briefly explain the physical significance of (i) Gauss's law for magnetism, and (ii) Faraday's law of electromagnetic induction in the context of these four equations. (c) How do these equations theoretically predict the existence of electromagnetic waves? [Practice] High

Q48. Case Study: Microwave Ovens Microwave ovens are common household appliances used for rapidly heating food. They utilize electromagnetic waves in the microwave frequency range. (a) State the exact principle of heating employed in a microwave oven. What is 'resonant frequency' in this context? (b) Why does a microwave oven heat food containing water much faster than dry food items? (c) Why is it strictly advised not to use metal containers inside a microwave oven? (d) Are the microwaves used in ovens harmful to human health if the door shielding is compromised? Why? [Practice] Very High

Q49. (a) Describe briefly the role of the Earth's atmosphere in selectively filtering electromagnetic waves. (b) What is the 'Greenhouse Effect'? Which specific electromagnetic waves are responsible for this, and how does the atmosphere interact with them? (c) How does the ionosphere facilitate long-distance short-wave radio broadcast? [Practice] Medium

Q50. (a) Deduce the relationship between the velocity of an electromagnetic wave in a vacuum (c), the permittivity of free space (ϵ_0), and the permeability of free space (μ_0). (b) An EM wave travels through a transparent liquid of refractive index 1.5. Calculate the speed of the wave in this liquid. (c) Prove mathematically that the total energy density of an electromagnetic wave is given by $u = \epsilon_0 E_{rms}^2$. [Practice] Very High

Answer Key

Q1: (d) Gamma rays. **Q2:** (a) electric field is changing. **Q3:** (c) ultraviolet rays. **Q4:** (a) X-rays. **Q5:** (c) microwave. **Q6:** (b) Ampere's circuital law. **Q7:** (d) Both electric and magnetic field vectors are parallel to each other. (They are mutually perpendicular). **Q8:** $1 : 1$. (The speed of all EM waves in a vacuum is $c = 3 \times 10^8\ \mathrm{m/s}$, regardless of wavelength). **Q9:** $0.25\ \mathrm{A}$. (Displacement current I_d is exactly equal to the conduction charging current I_c). **Q10:** It belongs to the Radio wave spectrum. It is used in radio astronomy to study the distribution of neutral hydrogen in the universe. **Q11:** $v = \frac{c}{\sqrt{\mu_r \epsilon_r}} = \frac{1}{\sqrt{\mu_0 \mu_r \epsilon_0 \epsilon_r}}$. **Q12:** The magnetic field $\vec{B}$ oscillates along the z-axis. (Since $\hat{E} \times \hat{B} = \hat{c} \Rightarrow \hat{j} \times \hat{k} = \hat{i}$). **Q13:** The ratio k/ω represents the inverse of the speed of the wave, $1/c$. **Q14:** Infrared rays. **Q15:** Ultraviolet rays. **Q16:** $[M^0 L^1 T^{-1}]$. (The ratio E_0/B_0 equals the speed of light c). **Q17:** (c) A is true but R is false. (In an EM wave, energy is divided equally between electric and magnetic fields). **Q18:** (b) Both A and R are true but R is NOT the correct explanation of A. (Displacement current flows whenever the electric field/potential changes, increasing or decreasing). **Q19:** (a) Both A and R are true and R is the correct explanation. **Q20:** (a) Both A and R are true and R is the correct explanation.

Q21: λ_1: Microwaves. λ_2: Ultraviolet rays. λ_3: Infrared rays. Ascending order: $\lambda_2 < \lambda_1 < \lambda_3$ (UV < Visible < IR < Microwave). **Q22:** During charging, the electric field between the plates increases with time, causing a change in electric flux. This changing flux produces a displacement current: $I_d = \epsilon_0 \frac{d\phi_E}{dt}$. **Q23:** Draw x, y, z axes. Draw a sine wave for $\vec{E}$ in the x-y plane and a synchronized sine wave for $\vec{B}$ in the x-z plane. Both must peak at the same points on the x-axis. **Q24:** It absorbs the majority of the Sun's harmful ultraviolet (UV) radiation, protecting DNA and living tissues from damage. **Q25:** (i) Microwaves ($10^9 - 10^{12}\ \mathrm{Hz}$). (ii) Ultraviolet rays ($10^{15} - 10^{16}\ \mathrm{Hz}$). **Q26:** Properties: Transverse nature, travel at speed of light in vacuum, don't require a medium. Momentum: They exert radiation pressure ($p = U/c$), causing a measurable force on objects. **Q27:** (a) X-rays; used in medical radiography. (b) Microwaves; used in radar/microwave ovens. **Q28:** An oscillating charge produces an oscillating electric field in space. This changing electric field produces an oscillating magnetic field, which in turn produces an oscillating electric field, and so on. Energy of the oscillating charge is converted into the energy of the radiated EM wave. **Q29:** They readily cause vibrations in molecules of matter, increasing their thermal energy. Produced by hot bodies and molecules. They trap the earth's outgoing radiation (Greenhouse effect), maintaining warmth. **Q30:** Produced by special vacuum tubes (klystrons/magnetrons). The frequency must match the resonant frequency of water so that maximum energy is efficiently transferred to the water molecules, generating heat. **Q31:** $u_E = \frac{1}{2}\epsilon_0 E_{rms}^2$. Since $E_{rms} = cB_{rms}$ and $c = 1/\sqrt{\mu_0 \epsilon_0}$, substituting gives $u_E = \frac{1}{2}\epsilon_0 (cB_{rms})^2 = \frac{1}{2}\epsilon_0 \frac{1}{\mu_0 \epsilon_0} B_{rms}^2 = \frac{B_{rms}^2}{2\mu_0} = u_B$. **Q32:** The magnitude of the electric field (E_0) and magnetic field (B_0) are related to the velocity v by the equation: $v = E_0/B_0$.

Q33: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_c + \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$. Inside the capacitor, $I_c = 0$. $E = \sigma/\epsilon_0 = q/(\epsilon_0 A)$. Flux $\phi_E = EA = q/\epsilon_0$. $d\phi_E/dt = (1/\epsilon_0)dq/dt = I/\epsilon_0 \Rightarrow I_d = \epsilon_0 d\phi_E/dt = I$. **Q34:** (a) $\vec{B}$ lies in the y-z plane (oscillates along the y-axis). (b) $\omega = 2\pi f = 2\pi \times 10^9 \text{ rad/s}$. $k = \omega/c = 2\pi \times 10^9/(3 \times 10^8) = \frac{20\pi}{3} \text{ m}^{-1}$. $B_0 = E_0/c = 50/(3 \times 10^8) = 1.67 \times 10^{-7} \text{ T}$. Equations: $E_x = 50 \sin(\frac{20\pi}{3} z - 2\pi \times 10^9 t)$ and $B_y = 1.67 \times 10^{-7} \sin(\frac{20\pi}{3} z - 2\pi \times 10^9 t)$. **Q35:** (a) Power $P = I \times A = (5 \times 10^{-14}) \times (\pi \times 0.2^2) = 6.28 \times 10^{-15} \text{ W}$. Energy $U = P \times t = 6.28 \times 10^{-15} \times (30 \times 60) = 1.13 \times 10^{-11} \text{ J}$. (b) Energy density $u = I/c = 5 \times 10^{-14}/(3 \times 10^8) = 1.67 \times 10^{-22} \text{ J/m}^3$. **Q36:** Area of hemisphere $A = 2\pi r^2 = 2\pi(10^5)^2 = 2\pi \times 10^{10} \text{ m}^2$. Intensity $I = P/A = 10^5/(2\pi \times 10^{10}) = 1.59 \times 10^{-6} \text{ W/m}^2$. $I = \frac{1}{2} c\epsilon_0 E_0^2 \Rightarrow E_0 = \sqrt{\frac{2I}{c\epsilon_0}} = 0.0346 \text{ V/m}$. $B_0 = E_0/c = 1.15 \times 10^{-10} \text{ T}$. **Q37:** Time for sound: $t_s = d/v = 2/340 = 5.88 \times 10^{-3} \text{ s}$. Time for EM wave: $t_{em} = D/c = 1765 \times 10^3/(3 \times 10^8) = 5.88 \times 10^{-3} \text{ s}$. Both events happen to take roughly the exact same time. **Q38:** Propagation along +x. $\vec{E}$ along -z, so $\vec{B}$ must be along +y (since $-\hat{k} \times \hat{j} = \hat{i}$). $\omega = 2\pi c/\lambda = 6\pi \times 10^{13} \text{ rad/s}$. $k = 2\pi/\lambda = 2\pi \times 10^5 \text{ m}^{-1}$. $B_0 = E_0/c = 10^6/(3 \times 10^8) = 3.33 \times 10^{-3} \text{ T}$. $\vec{E} = -10^6 \sin(2\pi \times 10^5 x - 6\pi \times 10^{13} t)\hat{k} \text{ V/m}$. $\vec{B} = 3.33 \times 10^{-3} \sin(2\pi \times 10^5 x - 6\pi \times 10^{13} t)\hat{j} \text{ T}$. **Q39:** Total $u = \epsilon_0 E_{rms}^2$. $E_{rms} = \sqrt{u/\epsilon_0} = \sqrt{4 \times 10^{-14}/8.85 \times 10^{-12}} = \sqrt{0.0045} = 0.067 \text{ V/m}$. Relationship: $E_{rms} = cB_{rms}$. **Q40:** Gamma Rays. Produced naturally by radioactive decay of atomic nuclei. Applications: Radiotherapy for destroying cancer cells; sterilizing surgical instruments. **Q41:** (a) Radiation pressure $P_r = F/A$. Force $F = dp/dt$. Momentum transferred $p = U/c$. Power $P = U/t$. $F = P/c$. So $P_r = P/(Ac) = I/c$. (b) No, Intensity I is power per unit area. If the identical uniform beam is merely intercepted by a smaller area, the power *received* drops proportionally, keeping the incident intensity constant. **Q42:** Ampere's law $\oint \vec{B} \cdot d\vec{l} = \mu_0 I$. If applied to a surface piercing the capacitor plates, $I = 0 \Rightarrow B = 0$, but a compass shows $B \neq 0$. Maxwell realized a changing electric field across the gap must act as a "displacement current" bridging the circuit.

Q43: (a) Current that comes into play in the region where the electric field and electric flux are changing with time. $I_d = \epsilon_0 d\phi_E/dt$. (b) By adding I_d to conduction current I_c, making the total current continuous. $\oint \vec{B} \cdot d\vec{l} = \mu_0(I_c + I_d)$. (c) Draw Amperian loop of radius r. $\oint B \cdot dl = B(2\pi r) = \mu_0 I_d$. Since flux is uniform, $I_d(r) = I\frac{\pi r^2}{\pi R^2}$. Thus $B = \frac{\mu_0 I r}{2\pi R^2}$. (d) No, once fully charged, E is constant, $dE/dt = 0$, so $I_d = 0$. **Q44:** (a) Microwaves ($10^9 - 10^{12} \text{ Hz}$). (b) X-Rays ($10^{16} - 10^{19} \text{ Hz}$). (c) Gamma rays ($> 10^{19} \text{ Hz}$). (d) Infrared waves ($10^{12} - 10^{14} \text{ Hz}$). (e) Visible light ($4 \times 10^{14} - 7.5 \times 10^{14} \text{ Hz}$). **Q45:** (a) E and B oscillate perpendicular to each other and perpendicular to the propagation direction. (b) They lie in the x-y plane. (c) The wave propagates along -x direction. (i) $k = 1.5 \times 10^3 \Rightarrow \lambda = 2\pi/k = 4.19 \times 10^{-3} \text{ m}$. (ii) $\omega = 4.5 \times 10^{11} \Rightarrow f = \omega/2\pi = 7.16 \times 10^{10} \text{ Hz}$. (iii) Since B is along z and wave is along -

x, E must be along -y. $E_y = -cB_0 \sin(kx + \omega t) = -60 \sin(1.5 \times 10^3 x + 4.5 \times 10^{11} t)\ \mathrm{V/m}$. **Q46:** (a) (i) $p = U/c$. (ii) $p = 2U/c$. (b) $F = P/c = 20/(3 \times 10^8) = 6.67 \times 10^{-8}\ \mathrm{N}$. (c) The force exerted is incredibly small (on the order of micronewtons over the entire body) due to the large denominator c. **Q47:** (a) Disturbances of mutually perpendicular E and B fields propagating through space. Equations: $\oint E \cdot dA = Q/\epsilon_0$; $\oint B \cdot dA = 0$; $\oint E \cdot dl = -d\phi_B/dt$; $\oint B \cdot dl = \mu_0(I_c + I_d)$. (b) (i) No magnetic monopoles. (ii) Changing magnetic field produces an electric field. (c) They show that a changing E-field creates a B-field, and a changing B-field creates an E-field, sustaining each other as a wave. **Q48:** (a) They emit microwaves matching the resonant frequency of water molecules, causing maximum amplitude of molecular vibrations and rapid heating. (b) Dry food lacks water molecules, so it doesn't absorb the microwaves efficiently. (c) Metals reflect microwaves, causing arcing, sparks, and potential damage to the oven's magnetron. (d) Yes, they can cause internal tissue burning due to their heating effect on body water. **Q49:** (a) The atmosphere blocks harmful X-rays/Gamma rays, absorbs most UV (via ozone) and IR (via greenhouse gases), but lets visible light and radio waves pass through "windows". (b) Trapping of outgoing IR radiation from the earth by CO_2/H_2O molecules, warming the planet. (c) The ionosphere reflects short-wave radio frequencies back to earth, allowing them to bounce over the horizon. **Q50:** (a) From Maxwell's equations, the speed of an EM wave is strictly given by $c = 1/\sqrt{\mu_0 \epsilon_0}$. (b) $v = c/n = (3 \times 10^8)/1.5 = 2 \times 10^8\ \mathrm{m/s}$. (c) $u = u_E + u_B = \frac{1}{2}\epsilon_0 E_{rms}^2 + \frac{1}{2\mu_0} B_{rms}^2$. Since $B_{rms} = E_{rms}/c$ and $1/\mu_0 = \epsilon_0 c^2$, the second term becomes $\frac{1}{2}\epsilon_0 E_{rms}^2$. Total $u = \epsilon_0 E_{rms}^2$.